

Double GREG estimation

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Problem setup

- ▶ Let A be a sample of size n selected from the population U of size N .
- ▶ Let π_i be the first order inclusion probability of unit i and π_{ij} the second order inclusion probability of unit i and j .
- ▶ Let y denote a variable of interest and we try to estimate the population total $t_y = \sum_{i \in U} y_i$.
- ▶ Let $\mathbf{x}_i$ of size p be the auxiliary variables associated with y . Assume that $\mathbf{x}_i$ is observed throughout the finite population while y_i is observed only in the sample.

Superpopulation model

- ▶ Suppose the following superpopulation model holds

$$y_i = \mathbf{x}_i^T \boldsymbol{\beta} + \varepsilon_i, \quad \varepsilon_i \stackrel{i.i.d.}{\sim} (0, \sigma^2), \quad i = 1, \dots, N \quad (1)$$

- ▶ Indeed, the linear regression model can be extended to the general nonlinear model whose basis function can be (penalized) spline functions. (1) can be replaced by

$$y_i = \sum_{j=0}^M h_j(\mathbf{x}_i) \beta_j + \varepsilon_i, \quad \varepsilon_i \stackrel{i.i.d.}{\sim} (0, \sigma^2), \quad i = 1, \dots, N \quad (2)$$

where $h_j(\cdot)$ are the corresponding basis functions. For the sake of simplicity, we assume linear basis functions in (1) only.

GREG estimator

- The working model (1) motivates the GREG estimator defined by

$$\hat{t}_y = \sum_{i \in U} \mathbf{x}_i^T \hat{\boldsymbol{\beta}} + \sum_{i \in A} \frac{1}{\pi_i} \left(y_i - \mathbf{x}_i^T \hat{\boldsymbol{\beta}} \right) \quad (3)$$

where $\hat{\boldsymbol{\beta}}$ is defined by

$$\hat{\boldsymbol{\beta}} = \left(\sum_{i \in A} \frac{1}{\pi_i} (\mathbf{x}_i - \bar{\mathbf{x}}) (\mathbf{x}_i - \bar{\mathbf{x}})^T \right)^{-1} \left(\sum_{i \in A} \frac{1}{\pi_i} (\mathbf{x}_i - \bar{\mathbf{x}}) y_i \right)$$

GREG estimator

(3) enjoys many desirable design properties including:

- ▶ asymptotic equivalence to the difference estimator
- ▶ design consistency
- ▶ asymptotically design unbiasedness
- ▶ asymptotically normally distribution with finite variance
- ▶ its design variance can consistently be estimated by the linearized variance estimator

$$\hat{V}_p = \sum_{i,j \in A} \frac{\Delta_{ij}}{\pi_{ij}} \frac{y_i - \mathbf{x}_i \hat{\boldsymbol{\beta}}}{\pi_i} \frac{y_j - \mathbf{x}_j \hat{\boldsymbol{\beta}}}{\pi_j} \quad (4)$$

However, when the number of parameters p grows as n increases, the properties listed above no longer hold.

High dimensional GREG estimator

- ▶ Now, suppose that p is allowed to increase as the sample size increases.
- ▶ In the high-dimension setup, the estimating errors of the regression coefficients $\hat{\beta}$ are no longer negligible.
- ▶ the GREG estimator is no longer asymptotically equivalent to the difference estimator since

$$\hat{t}_{y,GREG} - \hat{t}_{y,diff} = \left(\sum_{i \in U} \mathbf{x}_i - \sum_{i \in A} \frac{1}{\pi_i} \mathbf{x}_i \right) (\hat{\beta} - \beta) \neq o_p(n^{-1})$$

- ▶ Indeed, Ta et al. (2020) recently showed that the GREG estimator is asymptotically equivalent to the difference estimator only if $p/n \rightarrow 0$.

Bias of GREG estimator

- ▶ Under fixed p , the bias ($O(n^{-1})$) of the GREG estimator is negligible compared to its standard error ($O(n^{-1/2})$) (Cochran, 1977)
- ▶ On the other hand, the bias of the high dimensional GREG estimator can be significant.
- ▶ The simulation results show that such a tendency manifests itself even stronger when the sampling design is informative (in the next section).

Sample splitting

In this regard, we consider the sample split method. Let A_1 and A_2 be a random partition of A (possibly of equal size). Let $\hat{\beta}_1$ and $\hat{\beta}_2$ be the generalized regression coefficients using A_1 and A_2 respectively. We propose the double GREG estimator defined as follows

$$\hat{t}_y = \sum_{i \in U} \mathbf{x}_i^T \tilde{\beta} + \sum_{i \in A_1} d_i \left(y_i - \mathbf{x}_i^T \hat{\beta}_2 \right) + \sum_{i \in A_2} d_i \left(y_i - \mathbf{x}_i^T \hat{\beta}_1 \right) \quad (5)$$

where $\tilde{\beta}$ is the weighted regression coefficient defined by

$$\tilde{\beta} = \frac{\sum_{i \in A_2} d_i}{\sum_{i \in A} d_i} \hat{\beta}_1 + \frac{\sum_{i \in A_1} d_i}{\sum_{i \in A} d_i} \hat{\beta}_2 \quad (6)$$

Ensemble Learning

- ▶ Because of the random split, the resulting double GREG estimator can suffer from a large variance.
- ▶ In order to reduce the variance and improve efficiency, we take several random splits from a sample and take the average of all the samples.
- ▶ That is, for each of the random samples $A_1(1), \dots, A_1(K)$ and $A_2(1), \dots, A_2(K)$ the final ensemble estimator is defined as

$$\hat{t}_y = \frac{1}{K} \sum_{k=1}^K \hat{t}_y(A_1(k), A_2(k)) \quad (7)$$

Double GREG as linear estimator

- ▶ Observe that both (5) and (7) are a linear estimator of y .
- ▶ That is, the double estimator can be constructed as a weighted sum of y 's whose weights are not dependent on y . The proposed methods can be useful when multivariate y 's are used.
- ▶ If the design matrix $\mathbf{X}$ has an intercept, (5) is algebraically equivalent to

$$\hat{t}_y = \sum_{i \in U} \mathbf{x}_i^T \tilde{\beta} + \left(\sum_{i \in A_1} d_i \mathbf{x}_i - \sum_{i \in A_2} d_i \mathbf{x}_i \right)^T (\hat{\beta}_1 - \hat{\beta}_2)$$

Simulation setup

- ▶ A sample of size $n = 100$ is selected from the population of size $N = 2000$.
- ▶ $x_{ij} \sim N(2, 1)$
- ▶ Suppose the following super-population model holds

$$y_i = \mathbf{x}_i^T \boldsymbol{\beta} + \varepsilon_i, \quad \varepsilon_i \stackrel{i.i.d.}{\sim} N(0, \sigma^2), i = 1, \dots, N$$

where $\sigma^2 = 1$ and $\boldsymbol{\beta} = (1, \dots, 1, 0, \dots, 0)$ with $s = 20$ 1s and $p - s = 20$ 0s.

Simulation setup

We assume the samples are drawn from the informative stratified sampling.

$$Z_j \sim Z_j^* + \varepsilon_j, \quad Z^* \sim N\left(0, \frac{1-r}{r}\right), j \in U, r \in (0, 1)$$

The finite population data $\{y_j, \mathbf{x}_j, z_j\}_{j \in U}$ are sorted by z_j so that the 500 smallest z values are in stratum one and the next 500 smallest value are in stratum two and so forth. Within each stratum, simple random samples are collected with size

$$n_1 = 15, n_2 = 20, n_3 = 30, n_4 = 35$$

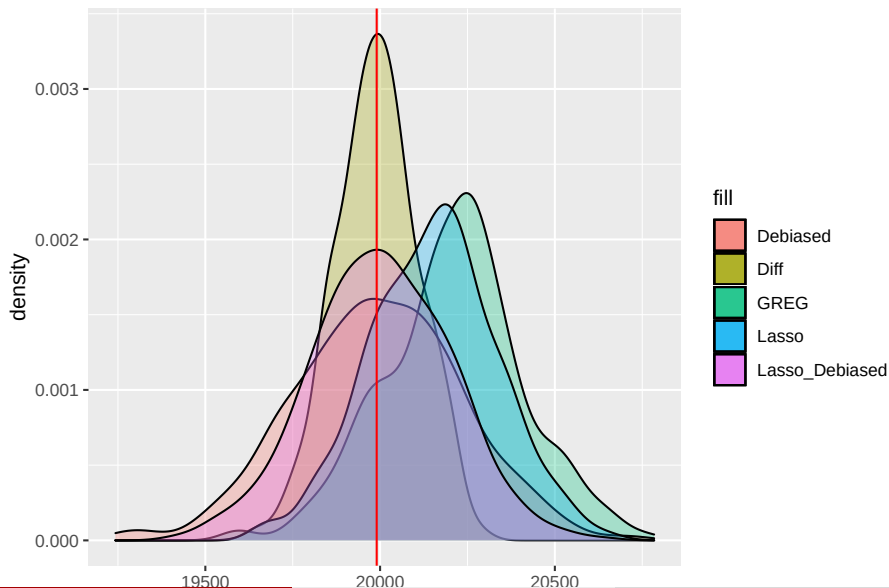
The larger r is, the more informative sampling design we have. We consider the following estimators. Here, we set $r = 0.75$.

Simulation results

	BIAS	SE	RMSE
HT	14.17	344.56	344.85
Diff	-0.08	118.29	118.29
GREG	218.85	198.73	295.62
Double_GREG	3.80	240.60	240.63
Lasso	161.60	181.59	243.08
Double_Lasso	12.39	198.25	198.64

Table:

Comparison of the estimators $\hat{t}_y$



References I